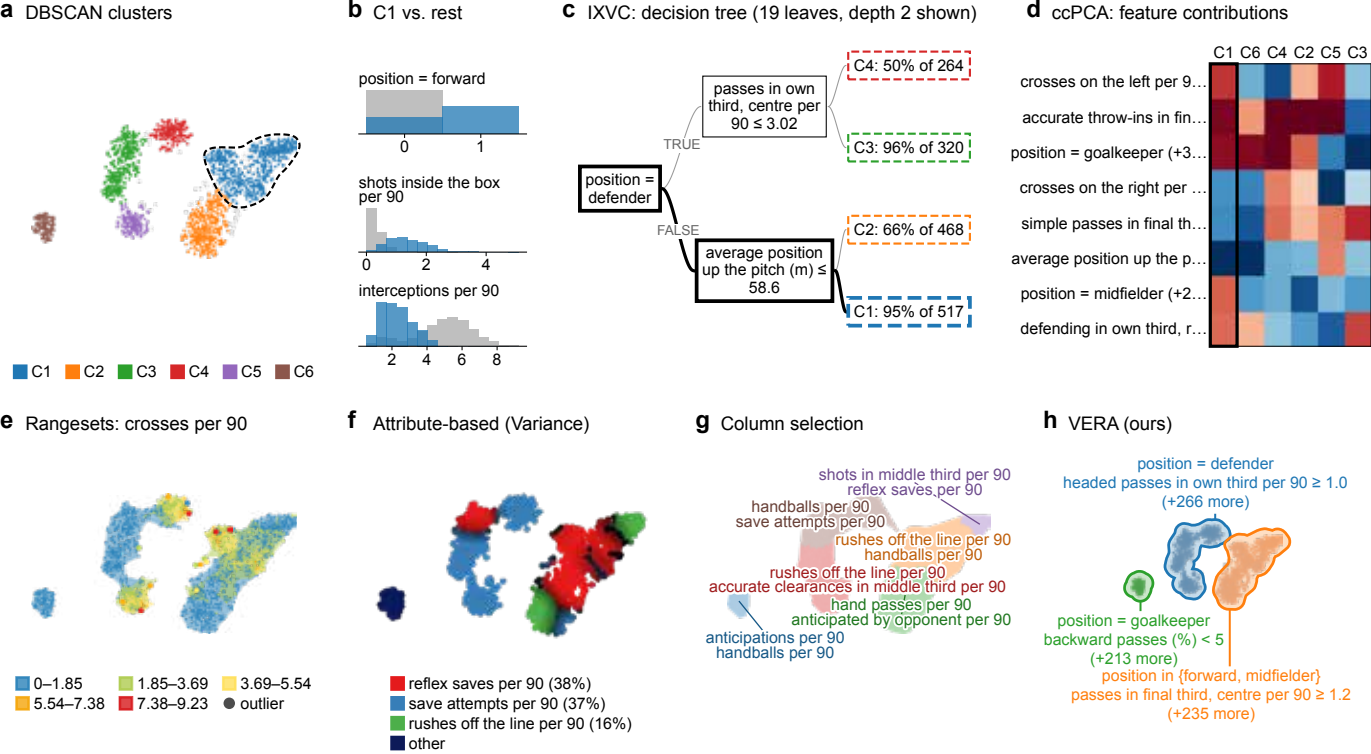

a DBSCAN clusters
C1
C2
C3
C4
C5
C6
b C1 vs. rest
position = forward
0
1
shots inside the box per 90
0
2
4
interceptions per 90
2
4
6
8
c IXVC: decision tree (19 leaves, depth 2 shown)
position = defender
TRUE
passes in own third, centre per 90 ≤ 3.02
C4: 50% of 264
C3: 96% of 320
FALSE
average position up the pitch (m) ≤ 58.6
C2: 66% of 468
C1: 95% of 517
d ccPCA: feature contributions
C1 C6 C4 C2 C5 C3
crosses on the left per 9…
accurate throw-ins in fin…
position = goalkeeper (+3…
crosses on the right per …
simple passes in final th…
average position up the p…
position = midfielder (+2…
defending in own third, r…
e Rangesets: crosses per 90
0–1.85
1.85–3.69
3.69–5.54
5.54–7.38
7.38–9.23
outlier
f Attribute-based (Variance)
reflex saves per 90 (38%)
save attempts per 90 (37%)
rushes off the line per 90 (16%)
other
g Column selection
shots in middle third per 90
reflex saves per 90
handballs per 90
save attempts per 90
rushes off the line per 90
handballs per 90
rushes off the line per 90
accurate clearances in middle third per 90
hand passes per 90
anticipated by opponent per 90
anticipations per 90
handballs per 90
h VERA (ours)
position = defender
headed passes in own third per 90 ≥ 1.0
(+266 more)
position = goalkeeper
backward passes (%) < 5
(+213 more)
position in {forward, midfielder}
passes in final third, centre per 90 ≥ 1.2
(+235 more)